SAMPLE FORM – NOT VALID FOR SUBMISSION TO BUILDING DEPARTMENTS

CERTIFICATE OF VERIFICATION

Note: This table completed by ~~HERS~~ECC Registry.

Project Name:	Enforcement Agency:
Dwelling Address:	Permit Number:
City and Zip Code:	Permit Application Date:

Title 24, Part 6, Section 160.2(b)2 **Ventilation and Indoor Air Quality for Attached Dwelling Units.** All dwelling units shall meet the requirements of ANSI/ASHRAE Standard 62.2-~~2019~~ 2022 Ventilation and Acceptable Indoor Air Quality in Low-Rise Residential Buildings subject to the amendments specified by Title 24, Part 6, Section 160.2(b)2

Title 24, Part 6, Section 160.2(b)2 **Ventilation and Indoor Air Quality for Attached Dwelling Units.** All dwelling units shall meet the requirements of ANSI/ASHRAE Standard 62.2-2019 Ventilation and Acceptable Indoor Air Quality in Low-Rise Residential Buildings subject to the amendments specified by Title 24, Part 6, Section 160.2(b)2

A. Whole-Dwelling Mechanical Ventilation - General Information

Note:

Non-dwelling units do not meet the definition for a dwelling unit as defined in Section 100.1(b). Non-dwelling units are not designed to provide independent living facilities and do not provide permanent provisions for living, sleeping, eating, cooking and sanitation.

01	Dwelling Unit Name	
02	Building Type	
03	Project Scope	
04	Total Conditioned Floor Area of Dwelling Unit (For addition projects the conditioned floor area equals existing area plus addition area)	
05	Number of Bedrooms in Dwelling Unit (For addition projects the number of bedrooms equals the existing bedrooms plus addition bedrooms)	
06	Ventilation System Type	
07	Ventilation Operation Schedule	
08	Fault Indicator Display (FID) Status	

~~**MCH-27b – Multifamily Ventilation**~~

B. Ventilation - Total Ventilation Rate

A mechanical supply system, exhaust system, or combination thereof shall provide whole-dwelling ventilation with outdoor air each hour at no less than the rate in 160.2(b)2Aiv

01	Total Required Ventilation rate, (Q_{tot})	

C. Installed Ventilation - Total Ventilation Rate

A mechanical supply system, exhaust system, or combination thereof shall provide whole-dwelling ventilation with outdoor air each hour at no less than the rate in 160.2(b)2Aiv

01	02	03	04	05
Fan Name	Fan Location	Runtime (Min/Hr)	Installed Mechanical Ventilation Rate (CFM)	Equivalent Continuous Ventilation (CFM)
06	Total Installed Equivalent Continuous Ventilation (CFM)			

~~C2~~D. HRV or ERV serving Individual Dwelling Unit

- Heat or Energy Recovery Systems must have a fan efficacy of ≤ 1.0 W/cfm in all climate zones (Section 160.2(b)2Biii).
- Heat or Energy Recovery Systems must prescriptively have a fan efficacy of ≤ 0.6 W/cfm and a minimum sensible heat recovery of 67% in climate zones 1, 2, 4~~and~~ 11-14 and 16 (Section 170.2(c)3Biva).

01	02	03	04
Manufacturer Make	Manufacturer Model Number	Fan Efficacy Performance Rating (W/CFM)	Sensible Recovery Efficiency (%)

~~D~~E. Additional Envelope Requirements

01	Envelope Leakage	

~~E~~F. Additional Central Ventilation System Balancing Requirements

01	Maximum Ventilation Flow (CFM)	

G. Fault Indicator Display Installation Verification

Qualification Requirements for Ventilation System Fault Indicator Displays are detailed in in Appendix JA17.

01	FID Manufacturer Name/Make	
02	FID Model Number	
03	The display module is mounted adjacent to the system thermostat.	
04	The manufacturer has certified to the Energy Commission that the FID model meets the requirements of Reference Joint Appendix JA17 (make and model found on CEC list of approved FID devices).	
05	The system has operated for at least 15 minutes and the FID reports that the system is operating within acceptable parameters.	
06	Compliance Statement:	

~~F~~H. Compliance Statement

01	

~~G~~I. Determination of ~~HERS~~ECC Verification Compliance

All applicable sections of this document shall indicate compliance with the specified verification protocol requirements in order for this Certificate of Verification as a whole to be determined to be in compliance

01	

DOCUMENTATION AUTHOR'S DECLARATION STATEMENT

1. I certify that this Certificate of Installation documentation is accurate and complete.

Documentation Author Name:	Documentation Author Signature:
Company:	Date Signed:
Address:	CEA/~~HERS~~ECC Certification Identification (If applicable):
City/State/Zip:	Phone:

RESPONSIBLE PERSON'S DECLARATION STATEMENT

I certify the following under penalty of perjury, under the laws of the State of California:

1. The information provided on this Certificate of Verification is true and correct.
2. I am the certified ~~HERS~~ECC Rater who performed the verification identified and reported on this Certificate of Verification (responsible rater).
3. The installed features, materials, components, manufactured devices, or system performance diagnostic results that require ~~HERS~~ECC verification identified on this Certificate of Verification comply with the applicable requirements in Reference Appendices RA2, RA3, and the requirements specified on the Certificate of Compliance for the building approved by the enforcement agency.
4. The information reported on applicable sections of the Certificate(s) of Installation (LMCI) signed and submitted by the person(s) responsible for the construction or installation conforms to the requirements specified on the Certificate(s) of Compliance (LMCC) approved by the enforcement agency.
5. I understand that a registered copy of this Certificate of Verification shall be posted, or made available with the building permit(s) issued for the building, and made available to the enforcement agency for all applicable inspections, and I will take the necessary steps to ensure this requirement is accomplished.
6. I understand that a registered copy of this Certificate of Verification is required to be included with the documentation the builder provides to the building.

Responsible Builder/Installer Name:	Responsible Builder/Installer Signature:	
Company Name: (Installing Subcontractor or General Contractor or Builder/Owner)	Position With Company (Title):	
Address:	CSLB License:	
City/State/Zip:	Phone:	Date Signed:
Third Party Quality Control Program (TPQCP) Status:	Name of TPQCP (if applicable):	

BUILDER OR INSTALLER INFORMATION AS SHOWN ON THE CERTIFICATE OF INSTALLATION

Company Name (Installing Subcontractor, General Contractor, or Builder/Owner):	
Responsible Builder or Installer Name:	CSLB License:

~~HERS~~ECC PROVIDER DATA REGISTRY INFORMATION

Sample Group Number (if applicable):	Dwelling Test Status in Sample Group (if applicable):

~~HERS~~ECC RATER INFORMATION

~~HERS~~ECC Rater Company Name:	
Responsible Rater Name:	Responsible Rater Signature:
Responsible Rater Certification Number w/ this ~~HERS~~ECC	Date Signed:

LMCV-MCH-27~~b~~-H User Instructions

Section A. General Information

1. Building Unit Name: This field is filled out automatically. It is referenced from the LMCI-MCH-01, which must be completed prior to this document. This is the unique identifier for this dwelling unit. Needed mostly for multifamily dwelling units. Ventilation is calculated and provided for each dwelling unit individually.
2. Building Type: This field is filled out automatically. It is referenced from the LMCC. Values are "Multifamily". User is allowed to overwrite imported value with "Non-dwelling unit" selection.
3. Project Scope: This field is filled out automatically. It is referenced from the LMCC.
   - If parent document is the LMCC-PRF-01, values are "Newly Constructed", "Newly Constructed (Addition Alone)" and "Addition and /or Alteration"
   - If parent document is CF1R-NCB-01, values are "Newly Constructed" and "Newly Constructed (Addition Alone)"
   - If parent document is CF1R-ADD-01, values are "ADU Addition < 300 ft^2"," ADU Addition > 300 to < 400 ft^2"," ADU Addition > 400 to < 700 ft^2" and "ADU Addition > 700 to < 1000 ft^2".
4. Total Conditioned Floor Area of Dwelling Unit: This field is filled out automatically. It is referenced from the LMCI-MCH-01.
5. Number of Bedrooms in Dwelling Unit: This field is filled out automatically. It is referenced from the LMCI-MCH-01.
6. Ventilation system Type: This may be filled out automatically or be user input.
   - If parent document is the LMCC-PRF-01, the value will be filled out automatically.
   - If building type is equal to Non-dwelling unit, an N/A value will be filled out automatically.
   - If parent document is the CF1R-NCB or CF1R-ADD, user selects from list of Supply, Exhaust, Balanced, Balanced – ERV, Balanced – HRV, Central Fan Integrated (CFI), Central Ventilation System – Supply and Central Ventilation System – Exhaust and Central Ventilation System Balanced.
7. Ventilation operation schedule: This may be filled out automatically or be user input.
   - If building type is equal to Non-dwelling unit, an N/A value will be filled out automatically.
   - User selects from list of Continuous, Short-Term Average, Scheduled and Real-time Control.
   - Note if "Ventilation System Type" (A11) = Central Fan Integrated & "Ventilation Operation Schedule" (A12) = Continuous; then user will not be allowed to proceed.

Section B. Whole Building Continuous Ventilation – Total Ventilation Rate Method

1. This value is automatically calculated using equation 160.2-B from the Energy Standards.

Section C. Installed Ventilation – Total Ventilation Rate Method

1. User input text identifying the fan name for each installed ventilation fan.
2. User input text identifying the fan location for each installed ventilation fan.
3. Runtime (Min/Hr): This value may be filled out automatically or be user input.
   - If ventilation operation schedule from section A = "continuous", then value of 60 will be automatically entered.
   - If ventilation operation schedule from section A = "short term average", then user enter value of less than or equal to 60 for each installed ventilation fan.
4. User to enter CFM value from test procedures described in RA3.7.4 for each installed ventilation fan.
5. Equivalent continuous ventilation CFM is automatically calculated for each ventilation fan.

6. Total installed equivalent continuous ventilation CFM is automatically calculated based on the installed ventilation fans.

Section ~~C2~~D. HRV or ERV serving Individual Dwelling Unit

1. User input manufacturer make of the installed equipment from the manufacturer nameplate.
2. User input model number of the installed equipment from the manufacturer nameplate.
3. User input the fan efficacy performance rating (W/CFM) for the installed equipment as determined by RA3.7.4.4.
4. User input the sensible recovery efficiency performance rating (%) for the installed equipment as determined by RA3.7.4.4.

Section ~~D~~E. Additional Envelope Requirements

1. Envelope Leakage: This field is filled out automatically. It is referenced from the LMCI-MCH-24, which must be completed prior to this document.

Section ~~E~~F. Additional Central Ventilation System Balancing Requirements

1. Maximum Ventilation Flow (CFM): This field is filled out automatically calculated.

Section G. Fault Indicator Display Installation Verification

1. Enter the manufacturer name or make of the approved Fault Indicator Display. Must match name shown on the list of approved devices kept by the Commission.
2. Enter the manufacturer model number of the approved Fault Indicator Display. Must match name shown on the list of approved devices kept by the Commission.
3. The installer must confirm that the FID display module is mounted adjacent to thermostat that controls the system being verified. This requirement is detailed in Residential Appendix JA17.
4. The installer must confirm that the installed FID is approved and appears the list of approved devices kept by the Commission. This requirement is detailed in Residential Appendix JA17.
5. The installer must confirm that the system has operated for at least 15 minutes and that they system is operating within acceptable parameters as specified by the FID and equipment manufacturers. This requirement is detailed in Residential Appendix JA17.

Section ~~F~~H. Compliance Statement

1. This field is filled out automatically.

Section ~~G~~I. Determination of ~~HERS~~ECC Verification Compliance

1. This field is filled out automatically based on all verification protocol requirements in this document showing compliance.

Title 24, Part 6, Section 160.2(b)2 **Ventilation and Indoor Air Quality for Attached Dwelling Units.** All dwelling units shall meet the requirements of ANSI/ASHRAE Standard62.2-20~~16~~22 Ventilation and Acceptable Indoor Air Quality in Low-Rise Residential Buildings subject to the amendments specified by Title 24, Part 6, Section 160.2(b)2A

A. Whole-Dwelling Mechanical Ventilation - General Information		
01	Dwelling Unit Name	<<calculated field, referenced data from MCH-01, "Dwelling Unit Name" (A01); else if information not available then allow user input.>>
02	Building Type	<< calculated field, referenced data from LMCC, allowed values = multifamily >>
03	Project Scope	<< calculated field, if parent is LMCC-PRF referenced data from LMCC-PRF; else if parent is LMCC-MCH allow user to select one from list. allowed values = Newly Constructed, Newly Constructed (Addition Alone), and Addition and /or Alteration; >>
04	Total Conditioned Floor Area of Dwelling Unit (For addition projects the conditioned floor area equals existing area plus addition area)	<<calculated field: referenced from MCH-01, "Dwelling Unit Total Conditioned Floor Area (ft^2)" (A03); else if information not available then allow user input >>
05	Number of Bedrooms in Dwelling Unit (For addition projects the number of bedrooms equals the existing bedrooms plus addition bedrooms)	<<calculated field: referenced from MCH-01, "Dwelling Unit Number of Bedrooms" (A09); if value from MCH-01 = 0 replace with 1; else if information not available then allow user input >>
06	Ventilation System Type	<< calculated value if parent is LMCC-PRF, reference data from LMCC-PRF; else if parent is LMCC-MCH, allow user to pick one from list: **Supply **Exhaust; or **Balanced; or **Balanced – ERV; or **Balanced – HRV; or **Central Fan Integrated (CFI); or **Central Ventilation System – Supply; or **Central Ventilation System – Exhaust; or **Central Ventilation System – Balanced Else if "Building Type" (A02) = "Non-dwelling unit" then value = N/A>>
07	Ventilation Operation Schedule	<< if "Building Type" (A02) = "Non-dwelling unit", then value = N/A; Else allow user to choose from list: **Continuous; or **Short-Term Average; Else if "Ventilation System Type" (A06) = Central Fan Integrated & "Ventilation Operation Schedule" (A07)= Continuous; then display: "Central Fan Integrated Ventilation System Not Allowed to Operate Continuously - **Do Not Proceed">>**
08	Fault Indicator Display (FID) Status	<<auto filled text: referenced from LMCI. Possible entries are "This system has a factory installed FID"; or "This system has a field installed FID"; or "This system does not have a FID device installed" If installed system does not match this entry, it can be overwritten by rater but it will be flagged as a possible fail>>>>
~~08~~	~~determine compliance method for this document; display applicable tables below;~~ ~~(this row not visible to user)~~	~~<<calculated field:~~ ~~"Building Type" (A02) = Multifamily and "Ventilation System Type" (A06) = Central Ventilation System – Supply, Central Ventilation System – Exhaust, or Central Ventilation System – Balanced, then display method:~~ ~~**27b – Multifamily**~~ ~~>>~~
09	Climate Zone (this row is not visible to the user)	<<value from LMCC>>
Note: Non-dwelling units do not meet the definition for a dwelling unit as defined in Section 100.1(b). Non-dwelling units are not designed to provide independent living facilities and do not provide permanent provisions for living, sleeping, eating, cooking and sanitation.		

~~**MCH-27b – Multifamily Ventilation**~~

B. Ventilation - Total Ventilation Rate

A mechanical supply system, exhaust system, or combination thereof shall provide whole- dwelling ventilation with outdoor air each hour at no less than the rate in 160.2(b)2Aiv

01	Total Required Ventilation rate, (Q_{tot})	<<calculated field, numeric: (use equation 150.0-B): [(0.03 * "Total Conditioned Floor Area of Dwelling Unit" (A04)) + (7.5*("Number of Bedrooms in Dwelling Unit" (A05) + 1)], (cfm). >>

C. Installed Ventilation - Total Ventilation Rate

A mechanical supply system, exhaust system, or combination thereof shall provide whole- dwelling ventilation with outdoor air each hour at no less than the rate in 160.2(b)2Aiv

01	02	03	04	05
Fan Name	Fan Location	Runtime (Min/Hr)	Installed Mechanical Ventilation Rate (CFM)	Equivalent Continuous Ventilation (CFM)
<< user input, text>>	<<user input, text>>	<<calculated field: if value in "Ventilation Operation Schedule" (B13) equals Continuous, then value equals 60; Else if value in "Ventilation Operation Schedule" (B13) equals Short Term Average ,then user input value positive integer ≤ 60>>	<< user input, positive integer>>	<<calculated field, value = ("Runtime (Min/Hr)" (C03) * "Installed Mechanical Ventilation Rate (CFM)" (C04)) / 60 (CFM)>>
06	Total Installed Equivalent Continuous Ventilation (CFM)			<<calculated field, value = sum of values in column "Equivalent Continuous Ventilation (CFM)" (C05)>>

~~C2~~D. HRV or ERV serving Individual Dwelling Unit

- Heat or Energy Recovery Systems must have a fan efficacy of ≤ 1.0 W/cfm in all climate zones (Section 160.2(b)2Biii).
- Heat or Energy Recovery Systems must prescriptively have a fan efficacy of ≤ 0.6 W/cfm and a minimum sensible heat recovery of 67% in climate zones 1, 2, and 11-16 (Section 170.2(c)3Biva).

01	02	03	04
Manufacturer Make	Manufacturer Model Number	Fan Efficacy Performance Rating (W/CFM)	Sensible Recovery Efficiency (%)
<< user input, text>>	<< user input, text>>	<< user input, text>> Range check: User entered value must be ≤ 0.6 if complying prescriptively in climate zones 1, 2, 4~~and~~ 11-14 and 16; else value must be ≤ 1.0>>	<< user input, text>> Range check: User entered value must be ≥ 67% if complying prescriptively in climate zones 1, 2, 4~~and~~ 11-14 and 16; else no restriction>>

~~D~~E. Additional Envelope Requirements

<<if Ventilation System Type (A06) equals Supply, Exhaust, Central Fan Integrated (CFI), Central Ventilation System – Supply or Central Ventilation System – Exhaust then display Table D; Else display the section does not apply message>>

01	Envelope Leakage	<< calculated field, referenced data from LMCI-MCH-24>>

~~E~~F. Additional Central Ventilation System Balancing Requirements

<<if Ventilation System Type (A06) equals Central Ventilation System – Supply, Central Ventilation System – Exhaust or Central Ventilation System – Balanced then display Table E; Else display the section does not apply message>>

01	Maximum Ventilation Flow (CFM)	<<calculated field, "Required Mechanical Ventilation Rate (B06) * 1.20>>

G. Fault Indicator Display Installation Verification

<<Procedures for the Fault Indicator Display Verification are detailed in in Appendix JA17>>

01	FID Manufacturer Name/Make	<<auto filled text: referenced from LMCI. If installed system does not match this entry, it can be overwritten by rater but it will be flagged as a possible fail.>>
02	FID Model Number	<<auto filled text: referenced from LMCI. If installed system does not match this entry, it can be overwritten by rater but it will be flagged as a possible fail.>>
03	The display module is mounted adjacent to the system thermostat.	<<user entry, select from choices "yes" or "no">>
04	The manufacturer has certified to the Energy Commission that the FID model meets the requirements of Reference Joint Appendix JA17 (make and model found on CEC list of approved FID devices).	<<user entry, select from choices "yes" or "no">>
05	The system has operated for at least 15 minutes and the FID reports that the system is operating within acceptable parameters.	<<user entry, select from choices "yes" or "no">>
06	Compliance Statement:	<<If A03, A04 and A05 = "yes", then print statement, "FID complies", Else print "FID does NOT comply>>

~~F~~H. Compliance Statement

01	<<If 'Building Type" (A02) = 'Non-dwelling Unit' then display text: "Building Passes"; Else if the "Ventilation System Type" (A06) equals Balanced, Balanced – ERV or Balanced – HRV and the "Total Installed Equivalent Continuous Ventilation" (C06) ≥ "Total Required Ventilation Rate" (B01), then display text: "Building Passes Mechanical Ventilation Rate Test" Else if the "Ventilation System Type" (A06) equals Central Ventilation System – Balanced and the "Total Installed Equivalent Continuous Ventilation" (C06) ≥ "Total Required Ventilation Rate" (B01) and the "Total Installed Equivalent Continuous Ventilation" (C06) ≤ "Maximum Ventilation Flow" (~~E~~F01), then display text: "Building Passes Mechanical Ventilation Rate Test" Else if the "Ventilation System Type" (A06) equals Supply, Exhaust, Central Fan Integrated (CFI) and the "Total Installed Equivalent Continuous Ventilation" (C06) ≥ "Total Required Ventilation Rate" (B01), and if "Envelope Leakage" (~~D~~E01) ≤ "Target dwelling unit compartmentalization leakage" (taken from LMCI-MCH-24), then display text: "Building Passes Mechanical Ventilation Rate Test" Else if the "Ventilation System Type" (A06) equals Central Ventilation System – Supply or Central Ventilation System – Exhaust and the "Total Installed Equivalent Continuous Ventilation" (C06) ≥ "Total Required Ventilation Rate" (B01), and if "Envelope Leakage"(~~D01~~E01) ≤ "Target dwelling unit compartmentalization leakage" (taken from LMCI-MCH-24), and the "Total Installed Equivalent Continuous Ventilation" (C06) ≤ "Maximum Ventilation Flow" (~~E~~F01), then display text: "Building Passes Mechanical Ventilation Rate Test" Else display text: "Building Fails Mechanical Ventilation Rate Test">>

~~G~~I. Determination of ~~HERS~~ECC Verification Compliance

All applicable sections of this document shall indicate compliance with the specified verification protocol requirements in order for this Certificate of Verification as a whole to be determined to be in compliance.

01	<<if ~~F~~H01 = Building Passes Mechanical Ventilation Rate Test, then display: Complies: All specified verification protocol requirements on this document are met; else display: Does not comply: One or more specified verification protocol requirements on this document are not met>>